

R Sai Pranav

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Profile Summary

Computer Science and Engineering student at REVA University with hands-on experience in full-stack web development, system design, and software engineering. Proficient in AI/ML engineering and backend development across multiple internships, with a filed patent, a national hackathon win, and ~46 merged open-source contributions to production infrastructure projects. Looking to apply strong problem-solving and engineering skills in a collaborative, product-focused environment.

Education

REVA University, Bachelor of Technology in Computer Science Engineering

2023 – Present

Bangalore, Karnataka | GPA: 8.54/10.0

Experience

Software Development Engineer (SDE) Intern, MES

Sep 2026 – Present

Cepheid India Pvt Ltd (Danaher Corporation) – Bangalore/Gurugram

- Develop and maintain Web APIs for integration on Opcenter (Siemens MES), customized for business needs, using SQL Server and AWS infrastructure.
- Work in an Agile model with monthly sprints across the full development lifecycle, from user stories to iterative deployment, maintaining code used in manufacturing today.

Software Engineering Intern, Full Stack

Dec 2025 – Aug 2026

MyStartupWave – Bangalore (Hybrid)

- Built and shipped 10+ full-stack web applications using React, FastAPI, Node.js, and MongoDB from scratch to production.
- Cut API response times by up to 35% through performance fixes and caching, backed by unit, integration, and automated API testing.

Project Intern, Cybersecurity Engineering

Nov 2025 – Aug 2026

iSPIRT Foundation (Sakram) – Remote

- Worked on protocol frameworks for India's Digital Public Infrastructure (DPI) using C++ and Python.
- Contributed to hardware-bound cryptographic validation and data integrity verification systems.

Research Intern, Explainable AI

Jul 2025 – Jan 2026

CSIR-4Pi (Fourth Paradigm Institute) – Remote

- Built a medical image analysis pipeline for breast cancer detection using MedSAM, ResNet, and U-Net in PyTorch.
- Added XAI heatmaps so doctors could understand model predictions; co-authoring a research paper on this work.

Open Source Contributions

- ~46 patches merged across PostgreSQL-ecosystem infrastructure projects (pgmoneta, pgagroal, pgvector — 18/14/7 respectively) and other C/Python codebases (GRASS GIS, JabRef, gnuradio, kubernetes/website), with 20+ additional PRs under active review.
- Found and fixed memory-safety defects in C — including a stack-buffer overflow from an unbounded strcat and multiple unchecked-realloc crashes — and root-caused an intermittent CI hang traced to an unbounded retry loop.
- Security fix (pending review) in interviewstreet/hiring-agent (7k★, HackerRank OSS): patched a prompt-injection vulnerability where PDF resumes could hide LLM-manipulating text by matching the page background color.

Projects & Achievements

OrbitClean — AWI SpaceTech Hackathon, Top 5

- Built a satellite image processing system to detect illegal dumps using Sentinel-2A imagery and a Random Forest classifier.
- Wrote a route optimization engine using the Clarke-Wright algorithm; built a risk predictor with AUC 0.80 over a 552-cell grid.

Aura AI Diagnostics — Cardano Hackathon Asia, National Winner (200+ teams)

- Built a healthcare diagnostics tool in Python that verifies patient data cryptographically without exposing raw records.
- Designed hash-map based data structures to handle patient records securely and efficiently.

HERB-Vault — SIH Blockchain Track

- Built a blockchain-based Ayurvedic herb traceability system using Hyperledger Fabric (2 orgs, Raft consensus) with Node.js chaincode.
- Developed a FastAPI + MongoDB backend, React 19 frontend, and offline-first React Native mobile app with SQLite, Zustand, and GPS zone enforcement.

Decentralized Spatial-Temporal Data Veracity System — Personal Patent, Filed 2026

- A decentralized system for spatial-temporal data veracity in distributed ledger environments with geospatial filtering, end-node credibility scoring, and cryptographic verification.

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, C++, SQL, HTML/CSS

Technologies: FastAPI, MongoDB, Docker, Git, Postman, Figma

Cloud: AWS, Google Cloud

Concepts: OOP, REST APIs, Microservices, System Design, CI/CD, Agile/Scrum, Data Structures & Algorithms, Observability & Logging

Leadership & Involvement

- Technical Lead, Algorand Club (REVA University, 2026–Present); organized 5+ hackathons, competed in 16+ incl. Google DeepMind, Cardano Asia, Adobe India, and HackRx; member, GDSC/GDG Bangalore and AWS Cloud Club (REVA University).